Third Year Engineering

21BTIS501-Operating System

Class - T.Y. (SEM-I)

Unit - IV

Memory Management

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Unit-IV Syllabus

- Background of Memory Management, Swapping, Contiguous Memory Allocation, Paging, Structure of the Page Table, Segmentation, Virtual Memory, Background, Demand Paging, Copy~on-Write, Page Replacement, Allocation of Frames, Thrashing.

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Why Memory Management is Required?

- Allocate and de-allocate memory before and after process execution.
- To keep track of used memory space by processes.
- To minimize fragmentation issues.
- To proper utilization of main memory.
- To maintain data integrity while executing of process.

What is Memory Management in an Operating System?

- It is technique of controlling and managing the functionality of RAM.
- It is used for achieving better concurrency, system performance, and memory utilization.
- Used to moves processes from primary memory to secondary memory and vice versa.
- To keeps track of available memory, memory allocation, and unallocated.

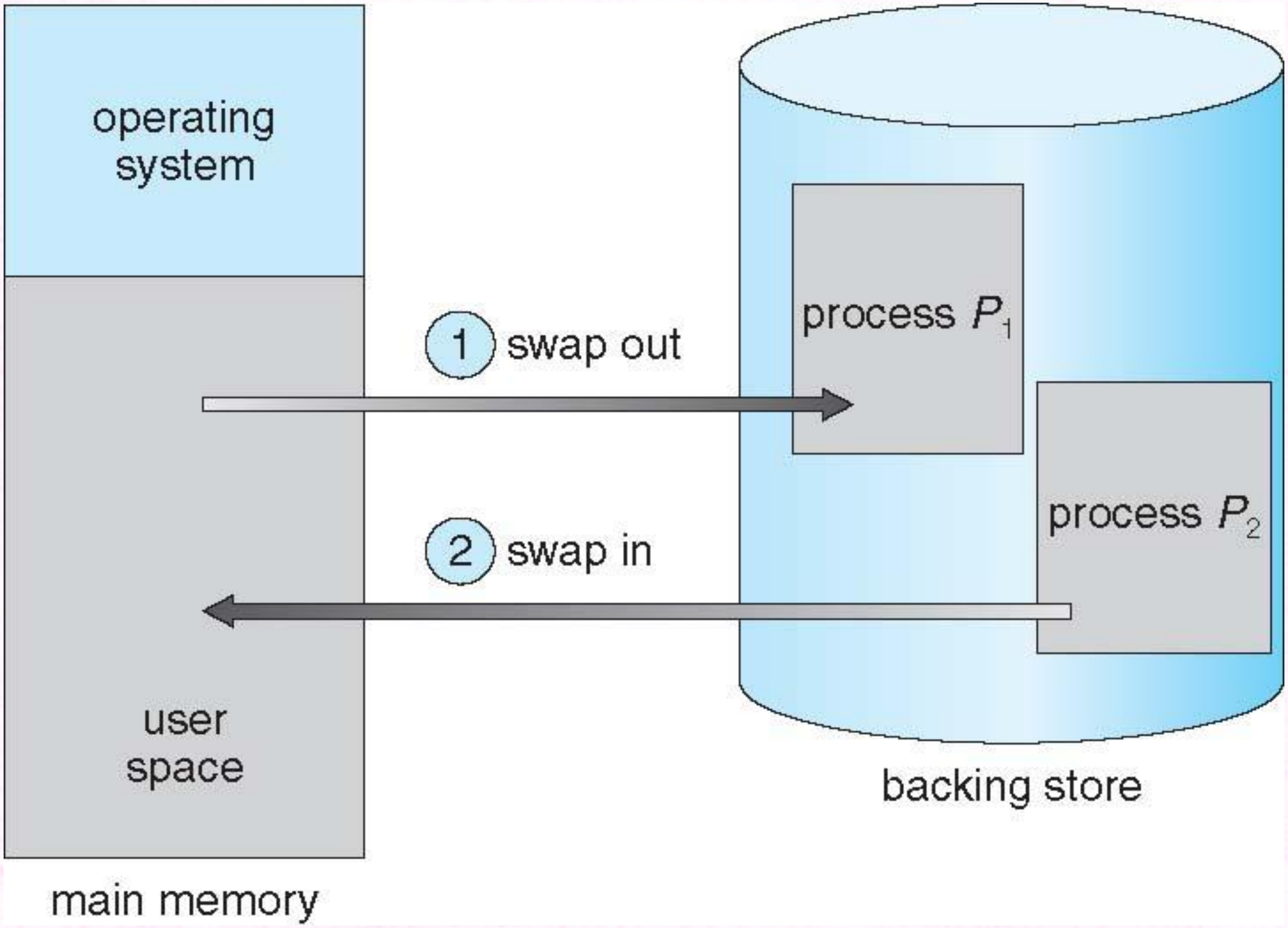
Swapping in an Operating System

- A process can be swapped temporarily out of memory to a backing store, and then brought back into memory for continued execution.
  - Total physical memory space of processes can exceed physical memory
- **Backing store** – fast disk large enough to accommodate copies of all memory images for all users; must provide direct access to these memory images
- **Roll out, roll in** – swapping variant used for priority-based scheduling algorithms; lower-priority process is swapped out so higher-priority process can be loaded and executed.

Swapping in an Operating System

- Major part of swap time is transfer time; total transfer time is directly proportional to the amount of memory swapped.
- System maintains a ready queue of ready-to-run processes which have memory images on disk.
- Modified versions of swapping are found on many systems (i.e., UNIX, Linux, and Windows)
  - Swapping normally disabled
  - Started if more than threshold amount of memory allocated
  - Disabled again once memory demand reduced below threshold

Schematic View of Swapping



Advantages of Swapping

- Efficient Memory Utilization.
- Better System Performance.
- Reduced Process Blocking.
- Flexibility.

Disadvantages of Swapping

- Performance Overhead
- Storage Overhead
- Data Integrity Issues
- Increased Disk Activity

Contiguous Memory Allocation

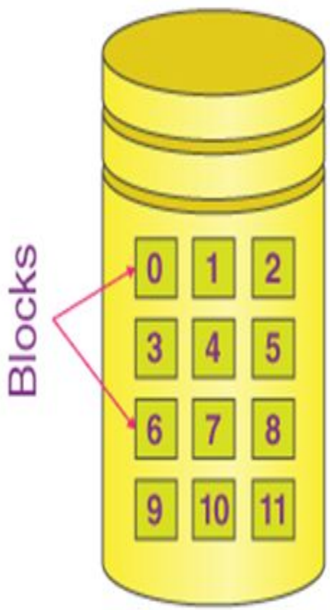
- Main memory must support both OS and user processes.
- Limited resource, must allocate efficiently
- Contiguous allocation is one early method
- Main memory usually into two partitions:
- Resident operating system, usually held in low memory with interrupt vector
- User processes then held in high memory.
- Each process contained in single contiguous section of memory

Contiguous Memory Allocation

- Relocation registers used to protect user processes from each other, and from changing operating-system code and data

1. Base register contains value of smallest physical address
2. Limit register contains range of logical addresses – each logical address must be less than the limit register
3. MMU maps logical address dynamically
4. Can then allow actions such as kernel code being transient and kernel changing size

Contiguous Memory Allocation



File Name	Start	Length	Allocated Blocks
abc.text	0	3	0, 1, 2
video.mp4	4	2	4, 5
jtp.docx	9	3	9, 10, 11

Directory

Contiguous Memory Allocation

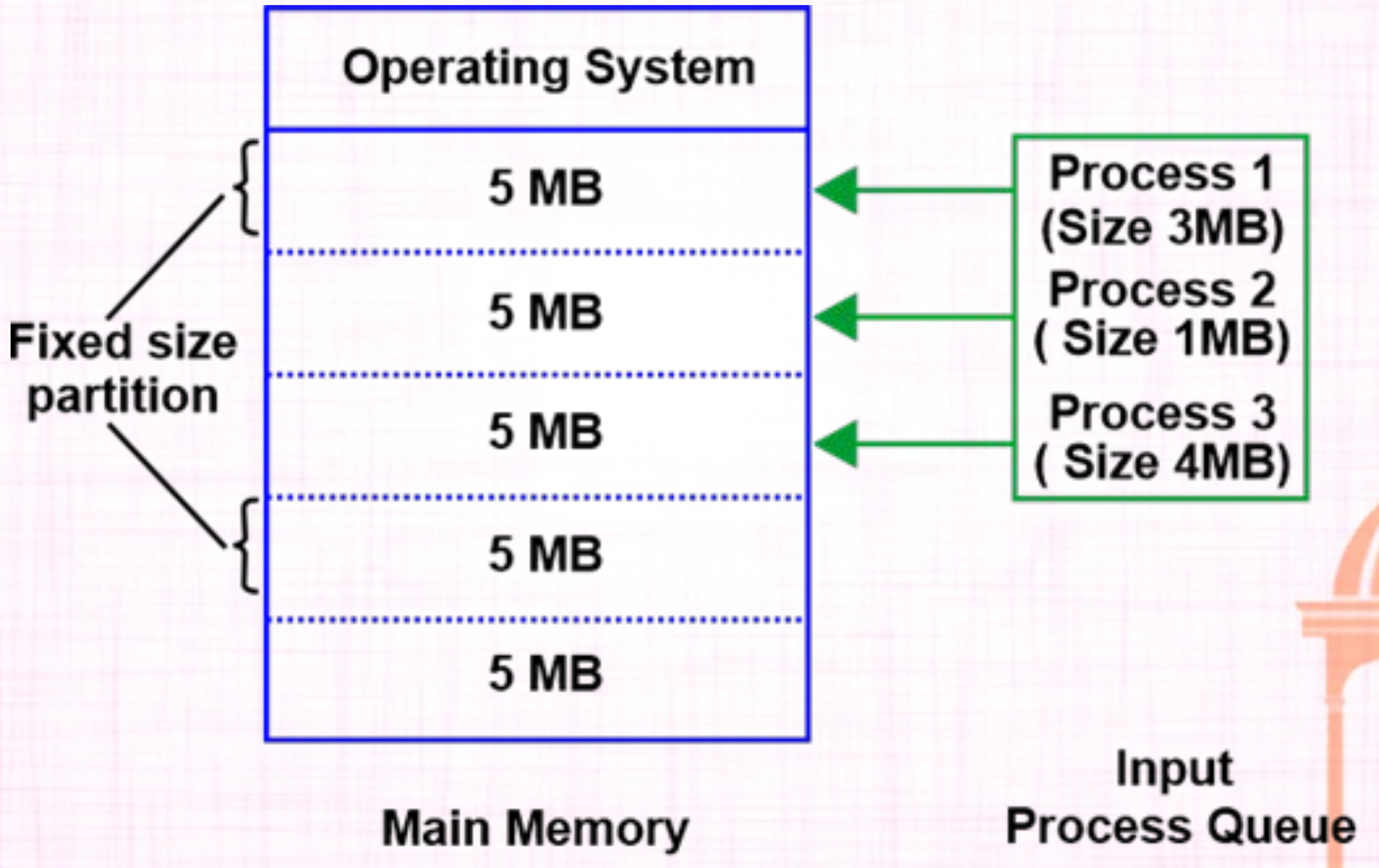
- This allocation can be done in two ways:

1. Fixed-size Partition Scheme
2. Variable-size Partition Scheme

1. Fixed-size Partition Scheme

- Another name for this is static partitioning. In this case, the system gets divided into multiple fixed-sized partitions.
- In this type of scheme, every partition may consist of exactly one process.
- This very process limits the extent at which multiprogramming would occur, since the total number of partitions decides the total number of processes.
- Read more on fixed-sized partitions here.

1. Fixed-size Partition Scheme



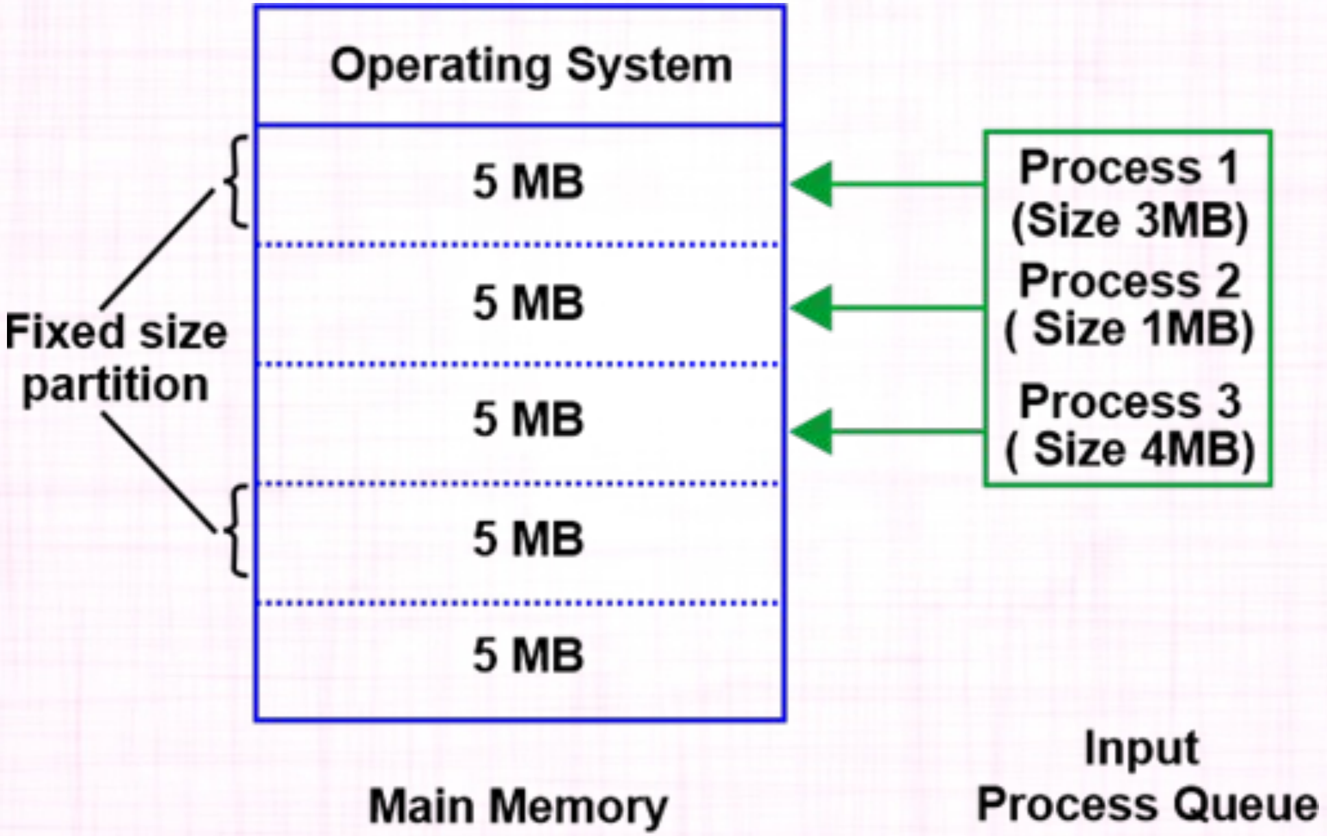
2. Variable-size Partition Scheme

- Dynamic partitioning is another name for this. The scheme allocation in this type of partition is done dynamically.
- Here, the size of every partition isn't declared initially.
- Only once we know the process size, will we know the size of the partitions.
- But in this case, the size of the process and the partition is equal; thus, it helps in preventing internal fragmentation.

2. Variable-size Partition Scheme

- On the other hand, when a process is smaller than its partition, some size of the partition gets wasted (internal fragmentation).
- It occurs in static partitioning, and dynamic partitioning solves this issue. Read more on dynamic partitions here.

2. Variable-size Partition Scheme



Memory Allocation

- Memory allocation is a process by which computer programs are assigned memory or space.
- Here, main memory is divided into two types of partitions

1. **Low Memory** – Operating system resides in this type of memory.
2. **High Memory**– User processes are held in high memory.

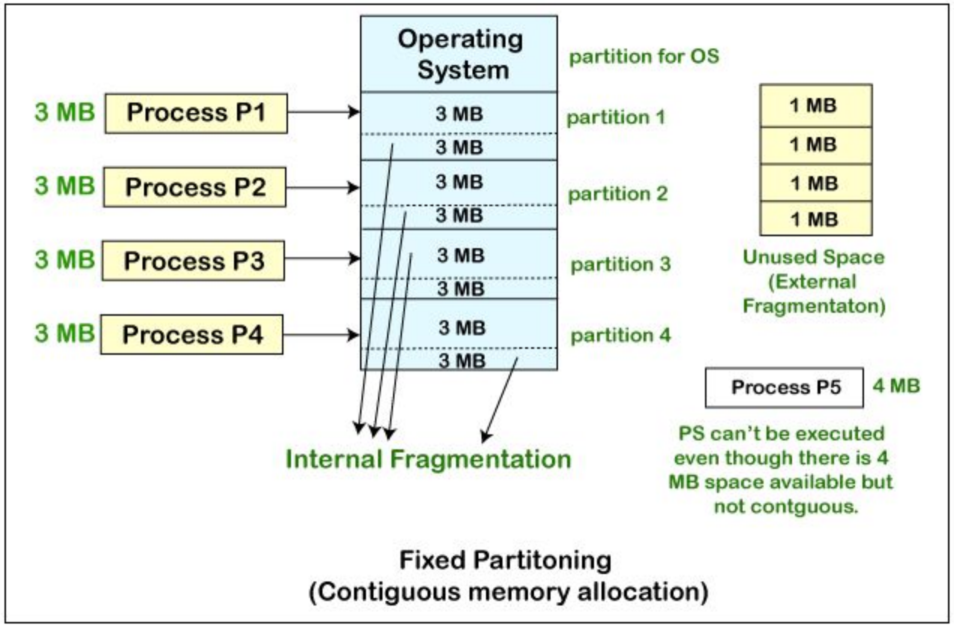
- Fixed Partitioning is also known as Contiguous memory allocation. Fixed Partitioning is the easiest method, which is used to load more than one process into the main memory.
- In Fixed Partitioning, we divide the main memory into partitions, and the size of partitions can be different or equal. In the first partition, the operating system is present, and the remaining partitions are used to store the user processes. In a contiguous way, we allocate the memory to the processes.
- In the fixed Partitioning:

1. There is no overlapping of partitions.
2. For the process execution, the process should be present contiguously.

Disadvantages of Fixed Partitioning

1. **Internal Fragmentation: –** In fixed partitioning, the partitions will be wasted and remain unused if the process size is lesser than the total size of the partition. In this way, memory gets wasted, and this is called **Internal fragmentation**.

- We can see in the fixed partitioning diagram that for loading a process of 3 MB, we use the partition of 4 MB, and in this way, 1 MB is getting wasted. Thus, this is the drawback of fixed partitioning.
- **External Fragmentation: –** We cannot use the total amount of unused space of different partitions to put the process even in the situation when we have some space available, but not in contiguous form. We can see in the fixed partitioning diagram, that we cannot use the 1 MB remaining space of each of the partition to store the process of 4 MB. Instead of that, we have a space available to load the process, but we cannot load the process. Such type of fragmentation is known as External Fragmentation.
- **Limitation on the Size of the Process: –** Sometimes, when the size of the process is larger than the maximum partition size, then we cannot load that process into the memory. So, this is the main disadvantage of the fixed partition.
- **Degree of Multiprogramming is Less: –** We can understand from the degree of multiprogramming that it means at the same time, the maximum number of processes we can load into the memory. In Fixed Partitioning, the size of the partition is fixed, and we cannot vary it according to the process size; therefore, in fixed partitioning, the degree of multiprogramming is less and fixed.


Operating System
partition for OS
3 MB
Process P1
3 MB
3 MB
partition 1
3 MB
Process P2
3 MB
3 MB
partition 2
3 MB
Process P3
3 MB
3 MB
partition 3
3 MB
Process P4
3 MB
3 MB
partition 4
1 MB
1 MB
1 MB
1 MB
Unused Space (External Fragmentaton)
Process P5
4 MB
PS can't be executed even though there is 4 MB space available but not contguous.
Internal Fragmentation
Fixed Partitoning
(Contiguous memory allocation)

Dynamic Partitioning

- Dynamic Partitioning is another technique for memory management that is invented to remove the problems that arises in the Fixed Partitioning technique.
- In dynamic partitioning, we do not declare the size of the partition in the starting. Instead, we declare the size of partition at the time of its loading.
- In this, the operating system, reserves the first partition. The rest of the other space is divided into different sections.
- The size of the partition and the size of the process remain equal.
- In dynamic partitioning, we can avoid the problem of internal fragmentation by varying the size of partition based on the needs of the process.

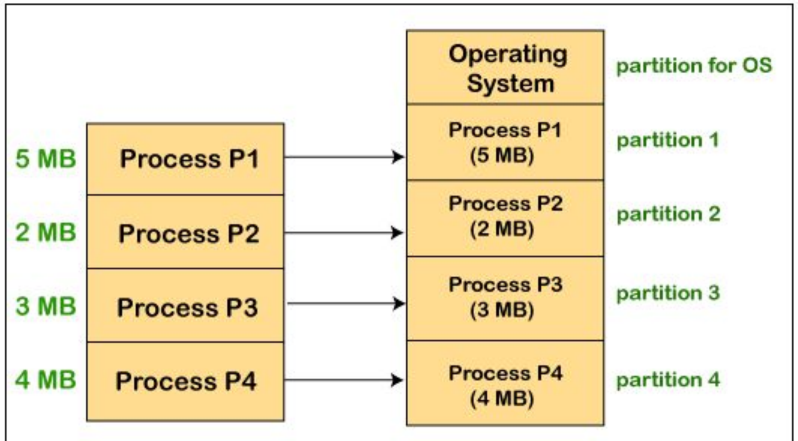
Advantages

1. **No Limitation on the size of the process:** – In fixed partitioning, if the size of the process is larger than the size of the partition, we cannot put or load the process into the memory. But if we talk about dynamic partitioning, the size of the process cannot be fixed, and we can change the size of partition according to the size of the process.
2. **The Degree of Multiprogramming is dynamic:** -In Dynamic Partition, there is no internal fragmentation, so there is no unused space that is present in the partition. So at the same time, we can load more number of processes in the memory.
3. **No Internal Fragmentation: –** In Dynamic Partitioning, the partitions are created dynamically as per the need for the process. So, in dynamic partitioning, internal fragmentation is not present, and the reason behind is that in dynamic partition, there is no space in the partition which remains unused.

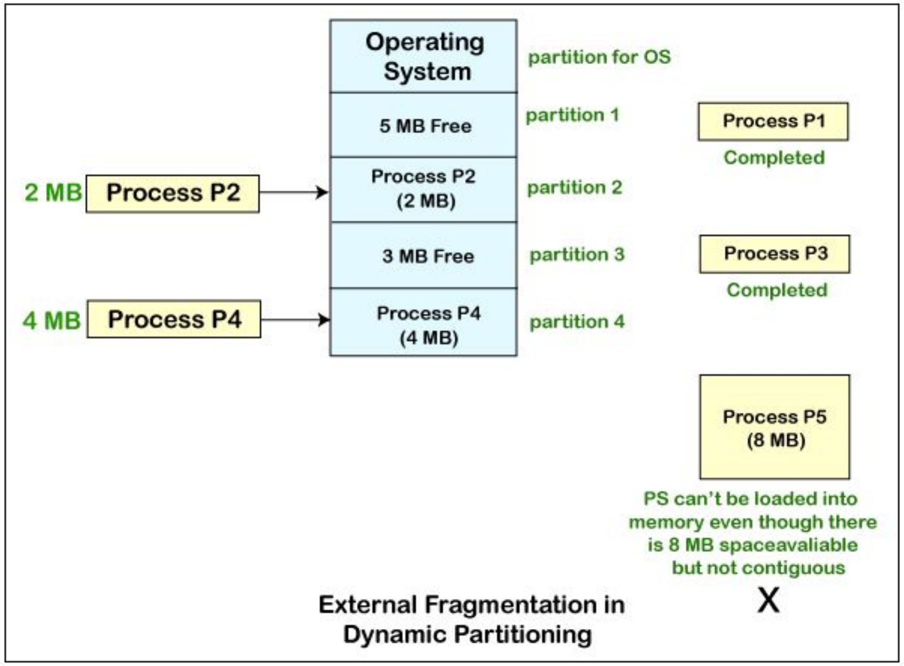
Disadvantages

1. **Complex Memory Allocation:** – In the case of fixed Partitioning, if once we create the list of partitions, we cannot modify it again. But in the case of dynamic partition, the task of allocation and deallocation is tough because the size of the partition varies whenever the partition is allocated to the new process. The operating system has to keep track of each of the partitions.

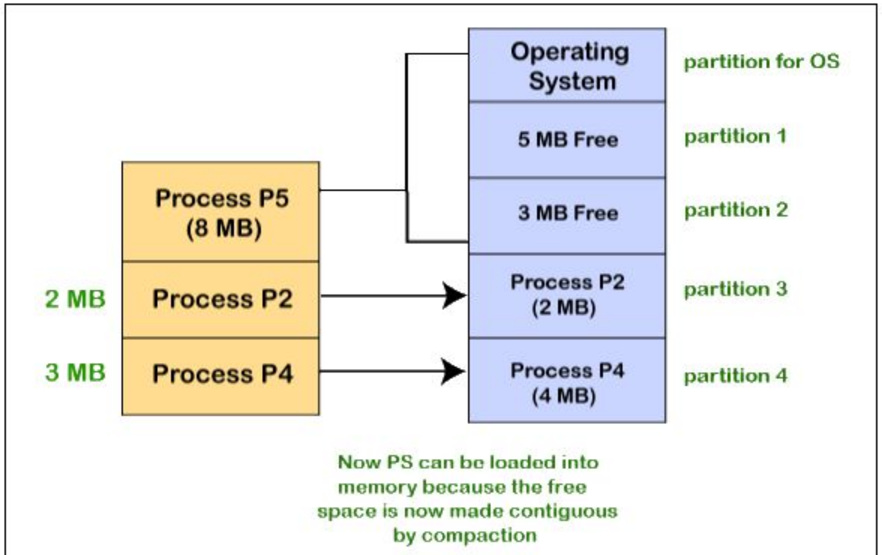
- So, due to the difficulty of allocation and deallocation in the dynamic memory allocation, and every time we have to change the size of the partition; therefore, it is tough for the operating system to handle everything.
- **External Fragmentation:** – One of the main disadvantages of dynamic partitioning is external fragmentation.
- Suppose we have three processes P1 (1MB), P2 (3MB), and P3 (1MB), and we want to load the processes in the various partitions of the main memory.
- Now the processes P1 and P3 are completed, and space that is assigned to the process P1 and P3 is freed. Now we have partitions of 1 MB each, which are unused and present in the main memory, but we cannot use this space to load the process of 2 MB in the memory because space is not contiguous.



Dynamic Partitioning
(Process Size = Partition Size)



Operating System
partition for OS
5 MB Free
partition 1
Process P1
Completed
2 MB
Process P2
Process P2 (2 MB)
partition 2
3 MB Free
partition 3
Process P3
Completed
4 MB
Process P4
Process P4 (4 MB)
partition 4
Process P5 (8 MB)
PS can't be loaded into memory even though there is 8 MB spaceavaliable but not contiguous
X
External Fragmentation in Dynamic Partitioning


Operating System
partition for OS
5 MB Free
partition 1
Process P5 (8 MB)
3 MB Free
partition 2
2 MB
Process P2
Process P2 (2 MB)
partition 3
3 MB
Process P4
Process P4 (4 MB)
partition 4
Now PS can be loaded into memory because the free space is now made contiguous by compaction

Compaction

Variable Size or Dynamic Partitions

Placement Algorithm

- Used to decide which free block to allocate to a process
- Goal: to reduce usage of compaction (time consuming)
- Possible algorithms:
  - Best-fit: choose smallest hole
  - First-fit: choose first hole from beginning
  - Next-fit: choose first hole from last placement

- **First Fit**: In this type fit, the partition is allocated, which is the first sufficient block from the beginning of the main memory.

- **Best Fit:** It allocates the process to the partition that is the first smallest partition among the free partitions.

- **Worst Fit:** It allocates the process to the partition, which is the largest sufficient freely available partition in the main memory.

- **Next Fit:** It is mostly similar to the first Fit, but this Fit, searches for the first sufficient partition from the last allocation point.

We have process p4 with 3kb requirement

OS
P1
<free> 10 KB
P2
<free> 16 KB
P3
<free> 4 KB

a. First fit algorithm allocates from the 10 KB block.
b. Best fit algorithm allocates from the 4 KB block.
c. Worst fit algorithm allocates from the 16 KB block.

New memory arrangements with respect to each algorithms will be as follows:

OS
P1
P4
<free> 7 KB
P2
<free> 16 KB
P3
<free> 4 KB

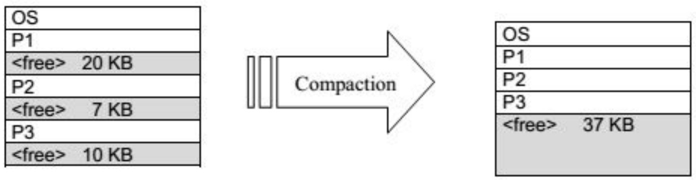
First Fit

OS
P1
<free> 10 KB
P2
<free> 16 KB
P3
P4
<free> 1 KB

Best Fit

OS
P1
<free> 10 KB
P2
P4
<free> 13 KB
P3
<free> 4 KB

Worst Fit


OS
P1
<free> 20 KB
P2
<free> 7 KB
P3
<free> 10 KB
Compaction
OS
P1
P2
P3
<free> 37 KB

Memory Allocation

- Memory allocation is a process by which computer programs are assigned memory or space.
- Here, main memory is divided into two types of partitions

1. **Low Memory** – Operating system resides in this type of memory.
2. **High Memory**– User processes are held in high memory.

Partition Allocation

- Memory is divided into different blocks or partitions.
- Each process is allocated according to the requirement.
- Partition allocation is an ideal method to avoid internal fragmentation.

Partition Allocation

- Below are the various partition allocation schemes :

1. First Fit: In this type fit, the partition is allocated, which is the first sufficient block from the beginning of the main memory.
2. Best Fit: It allocates the process to the partition that is the first smallest partition among the free partitions.

Partition Allocation

3. Worst Fit: It allocates the process to the partition, which is the largest sufficient freely available partition in the main memory.
4. Next Fit: It is mostly similar to the first Fit, but this Fit, searches for the first sufficient partition from the last allocation point.

Advantages of Contiguous Memory Allocation

1. It supports a user's random access to files.
2. The user gets excellent read performance.
3. It is fairly simple to implement.

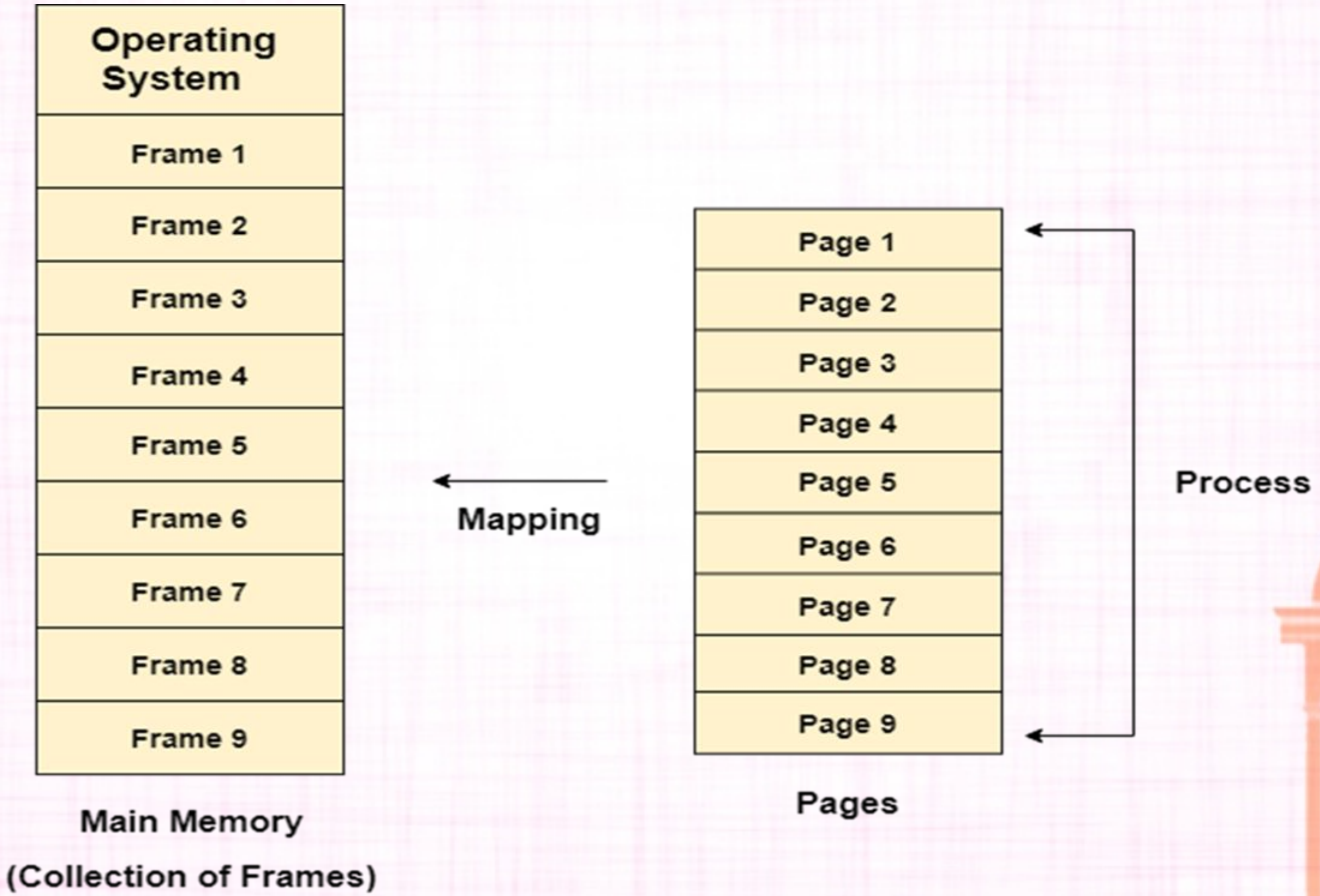
Disadvantages of Contiguous Memory Allocation

1. Having a file grow might be somewhat difficult.
2. The disk may become fragmented.

Paging in Operating System

- Logical address space of a process can be noncontiguous; process is allocated physical memory whenever the latter is available.
- Divide physical memory into fixed-sized blocks called frames (size is power of 2, between 512 bytes and 8192 bytes).
- Divide logical memory into blocks of same size called pages.
- Keep track of all free frames.
- To run a program of size n pages, need to find n free frames and load program.
- Set up a page table to translate logical to physical addresses.
- Internal fragmentation.

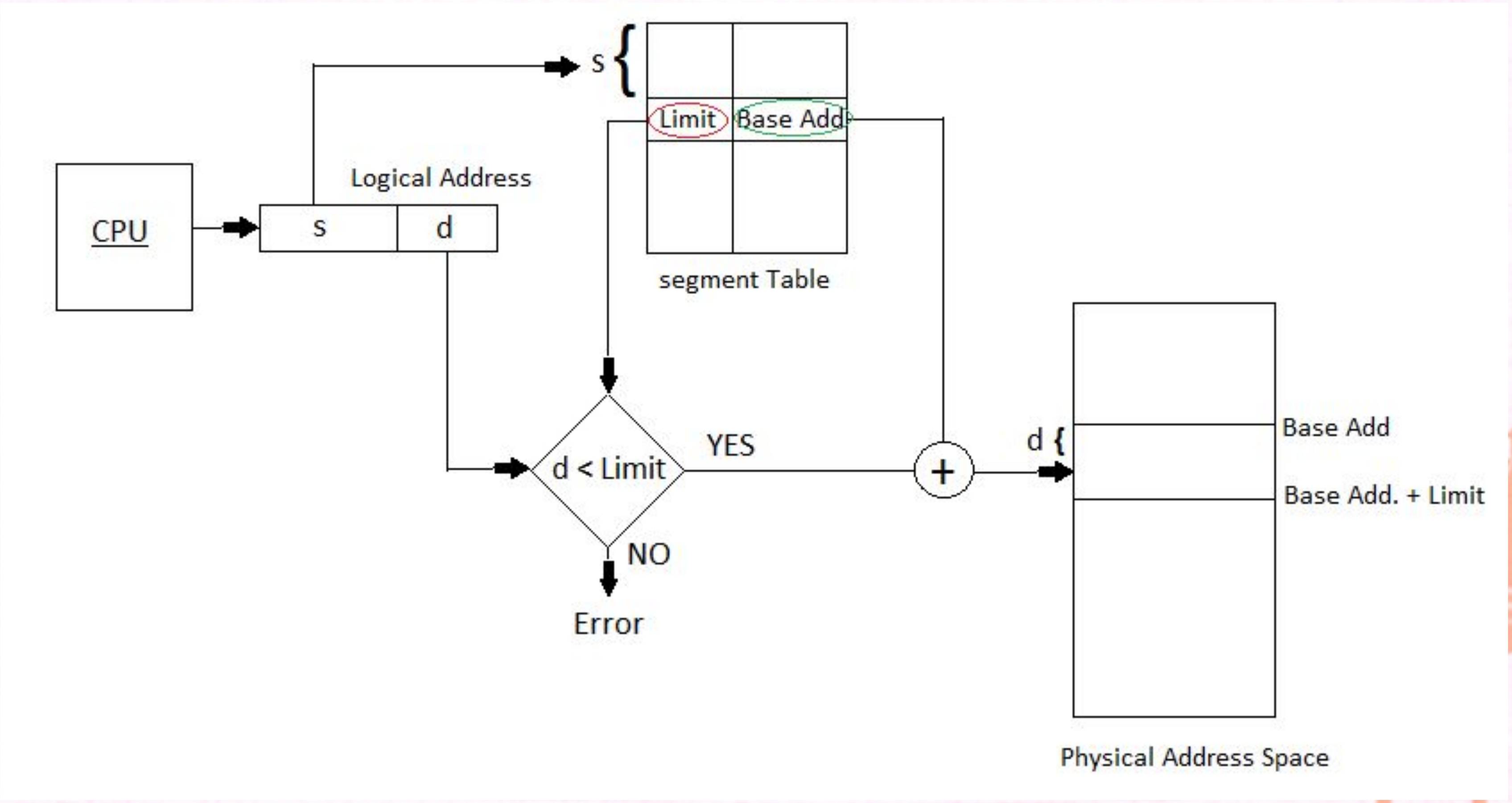
Paging in Operating System



Segmentation in Operating System

- Segmentation is a technique to break memory into logical pieces where each piece represents a group of related information.
- Segmentation is one of the most common ways to achieve memory protection
- Because internal fragmentation of pages takes place ,the user's view of memory is lost
- The user will view the memory as a combination of segments • In this type, memory address used are not contiguous.
- Each memory segment is associated with specific length and set of permission

Segmentation in Operating System



Segmentation in Operating System

- When a process tries to access the memory it is first checked to see whether it has the required permission to access the particular memory segment by the particular memory.
- The details about each segment are stored in a table called a segment table.
- Segment table is stored in one (or many) of the segments. Segment table contains mainly two information about segment:
    1. Base: It is the base address of the segment
    2. Limit: It is the length of the segment.

Difference between Paging and Segmentation

Paging	Segmentation
Main memory is partitioned into frames or blocks	Main memory is partationed into segments
The logical address space is divided into pages by MMU	The logical address space is divided into segments as specified by the program
The scheme suffers from internal fragmentation or page breaks	The scheme suffers from External fragmentation
OS maintains a free frame list , so that searching of free frame is not necessary	OS maintains the particulars of available memory
OS maintains a page map table for mapping between frames and pages	OS maintains a segment map table for mapping
This scheme does not support the users view of memory	This scheme support the users view of memory
Processor use the page number and displacement to calculate absolute address (p, d)	Processor use the segment number and displacement to calculate the absolute address (p, d)

Need of Virtual Memory

- In case, if a computer running the Windows operating system needs more memory or RAM than the memory installed in the system then it uses a small portion of the hard drive for this purpose.
- Suppose there is a situation when your computer does not have space in the physical memory, then it writes things that it needs to remember into the hard disk in a swap file and that as virtual memory.

Disadvantages of Virtual Memory

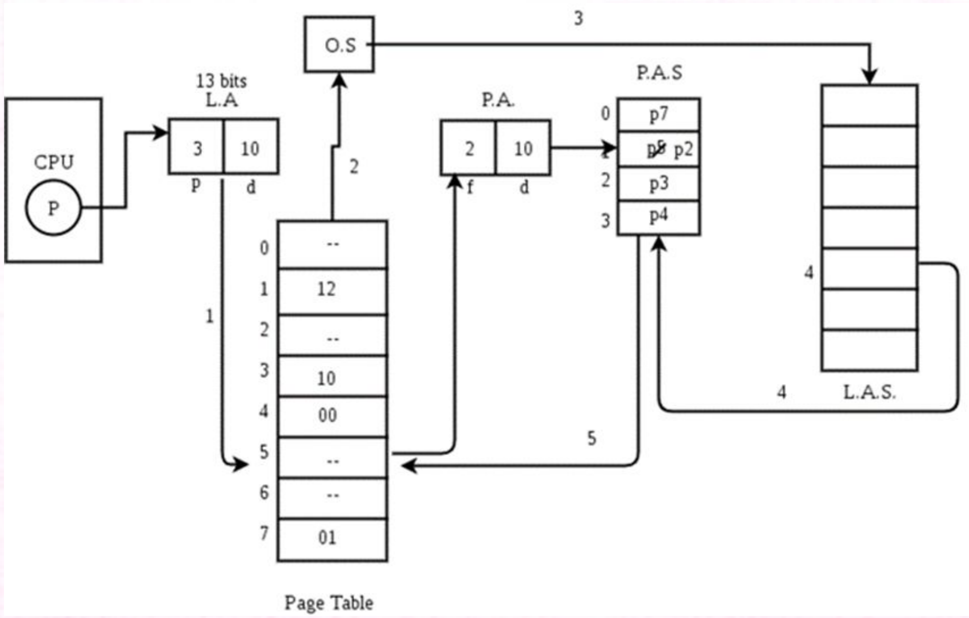
- Special hardware for address translation - some instructions may require 5-6 address translations!
- Difficulties in restarting instructions (chip/microcode complexity).
- Complexity of OS.
- Overhead - a Page-fault is an expensive operation in terms of both CPU and I/O overhead.
- Page-faults bad for real time.
- Thrashing problem

Advantages of Virtual Memory

- It can handle twice as many addresses as main memory.
- It enables more applications to be used at once.
- It has increased security because of memory isolation.
- It enables multiple larger applications to run simultaneously.
- Allocating memory is relatively inexpensive.
- It does not need external fragmentation.
- Data can be moved automatically.

Demand Paging in Operating System

- The process of loading the page into memory on demand (whenever a page fault occurs) is known as demand paging.
- The process includes the following steps are as follows:



Demand Paging in Operating System

- If the CPU tries to refer to a page that is currently not available in the main memory, it generates an interrupt indicating a memory access fault.
- The OS puts the interrupted process in a blocking state. For the execution to proceed the OS must bring the required page into the memory.
- The OS will search for the required page in the logical address space.
- The required page will be brought from logical address space to physical address space.

Demand Paging in Operating System

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- The required page will be brought from logical address space to physical address space.

Advantages of Demand Paging in OS

1. Memory can be put to better use.
2. If we use demand paging, then we can have a large virtual memory.
3. By using demand paging, we can run programs that are larger than physical memory.
4. In demand paging, there is no requirement for compaction.
5. In demand paging, the sharing of pages is easy.
6. Partition management is simple in demand paging because of the fixed partition size and the discontinuous loading.

Disadvantages of Demand Paging in OS

1. Internal fragmentation is a possibility with demand paging.
2. It takes longer to access memory (page table lookup).
3. Memory requirements
4. Guarded page tables
5. Inverted page tables

Copy on Write in Operating System

- Copy on Write or simply COW is a resource management technique. One of its main use is in the implementation of the fork system call in which it shares the virtual memory(pages) of the OS.
- In UNIX like OS, fork() system call creates a duplicate process of the parent process which is called as the child process.

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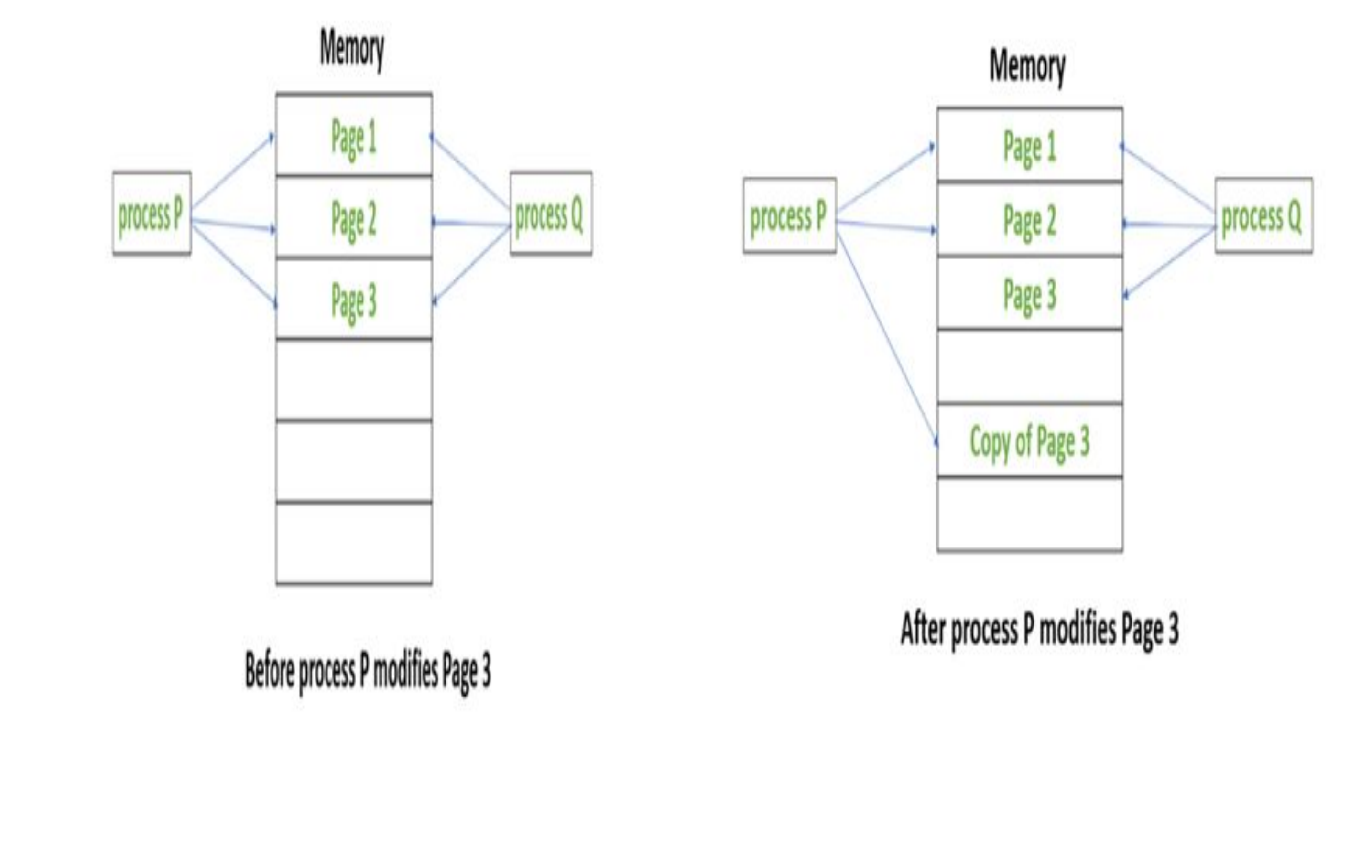
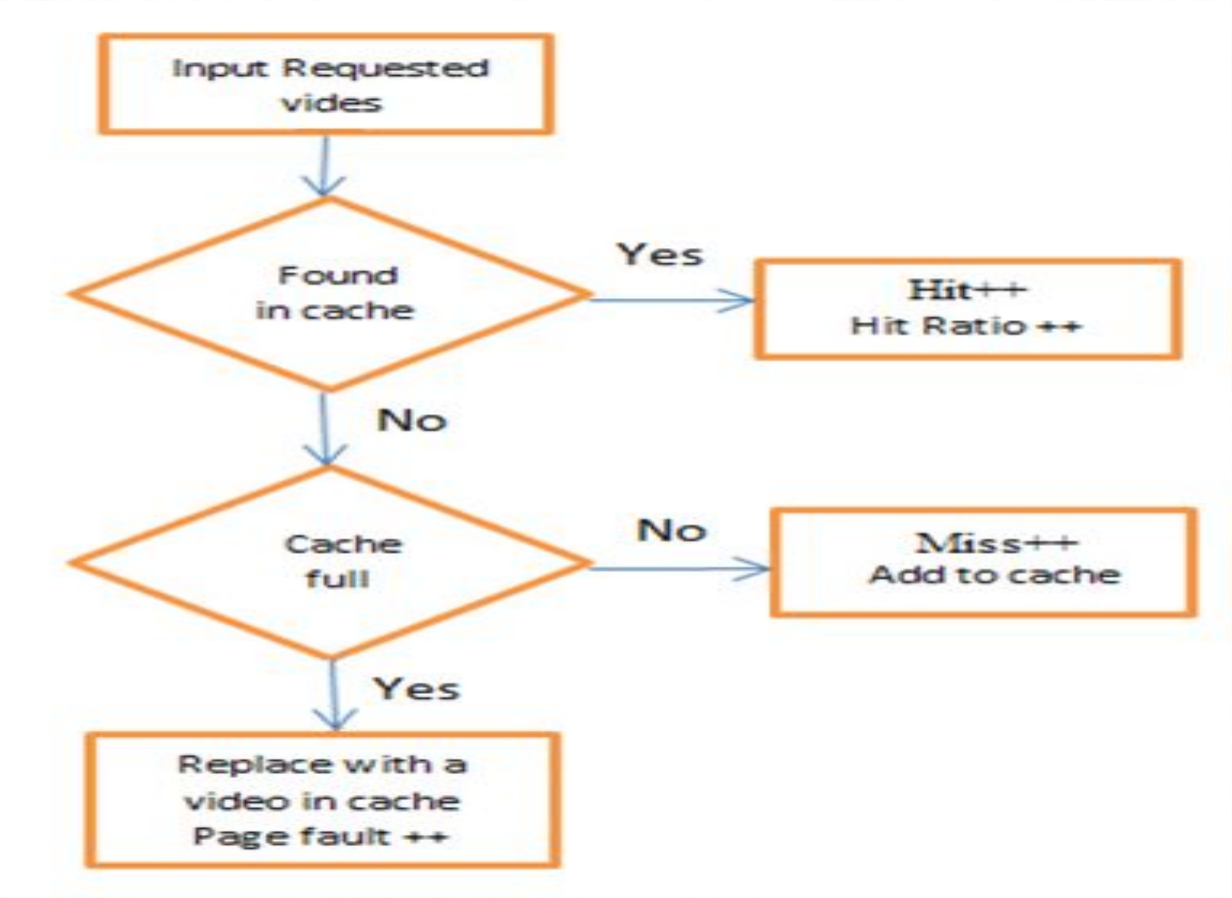
Copy on Write in Operating System



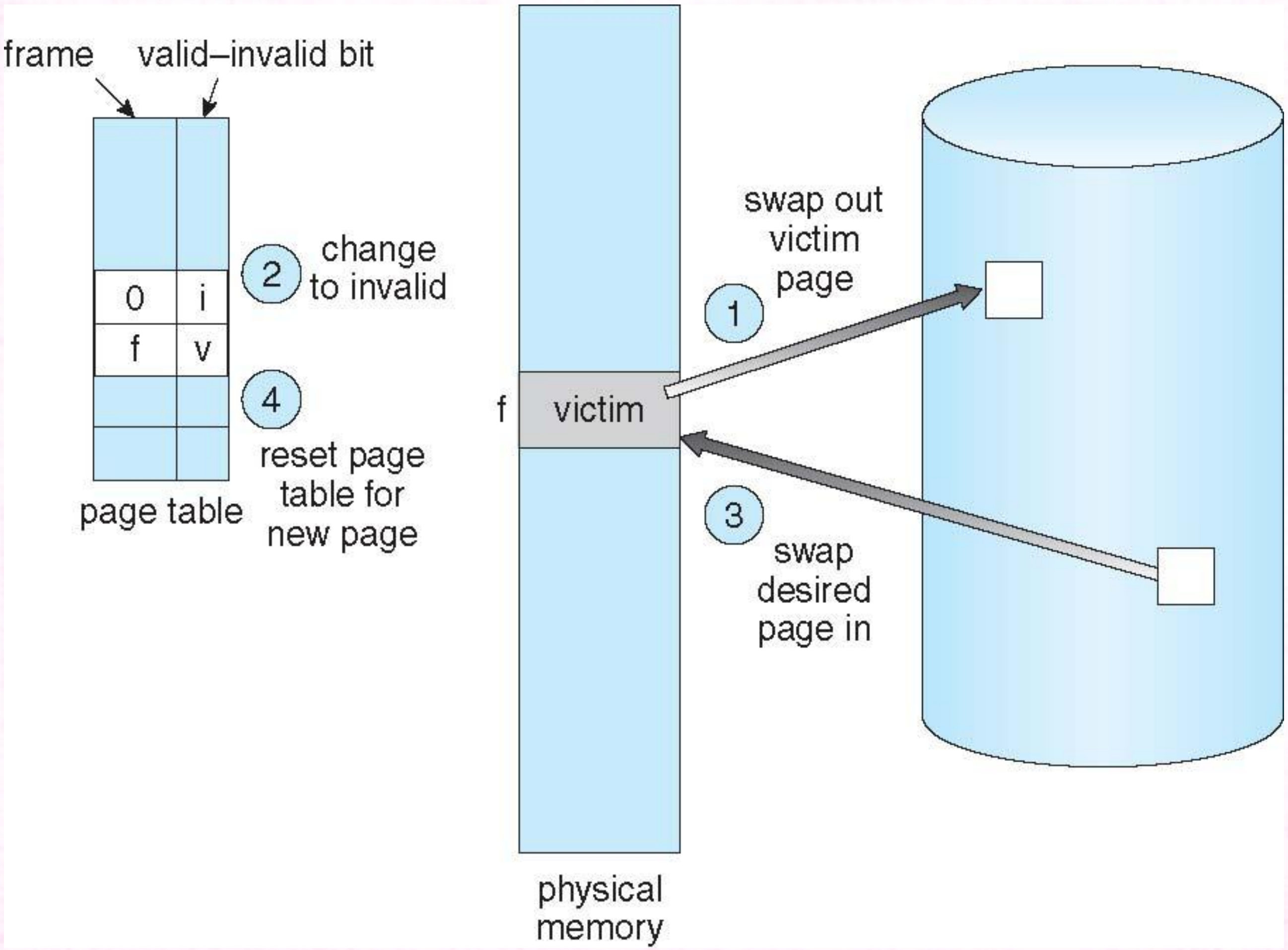
Page Replacement in Operating System

- Prevent over-allocation of memory by modifying page-fault service routine to include page replacement
- Use modify (dirty) bit to reduce overhead of page transfers – only modified pages are written to disk
- Page replacement completes separation between logical memory and physical memory – large virtual memory can be provided on a smaller physical memory

Page Replacement Algorithm Workflow



Basic Page Replacement



Page Replacement in Operating System

- Page Replacement Algorithms in Operating Systems

1. First In First Out (FIFO)
2. Least Recently Used (LRU)
3. Optimal Page Replacement

First-In-First-Out (FIFO) Algorithm

- Reference string: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5
- 3 frames (3 pages can be in memory at a time per process)

1	1	4	5	
2	2	1	3	9 page faults
3	3	2	4	

- 4 frames

1	1	5	4	
2	2	1	5	10 page faults
3	3	2		
4	4	3		

- Belady's Anomaly: more frames ⇒ more page faults

First-In-First-Out (FIFO) Algorithm

Reference string: 1, 2, 4, 7, 1, 2, 9, 1, 2, 4

- Use 2 frames:

1	2	4	7	1	2	9	1	2	4
1	2	4	7	1	2	9	1	2	4
	1	2	4	7	1	2	9	1	2

- # of page faults = 10
- Use 3 frames:

1	2	4	7	1	2	9	1	2	4
1	2	4	7	1	2	9	9	9	4
	1	2	4	7	1	2	2	2	9
		1	2	4	7	1	1	1	2
F	F	F	F	F	F	F			F

- # of page faults = 8

First-In-First-Out (FIFO) Algorithm

Reference string: 1, 2, 4, 7, 1, 2, 9, 1, 2, 4

- Use 4 frames:

1	2	4	7	1	2	9	1	2	4
1	2	4	7	7	7	9	1	2	4
	1	2	4	4	4	7	9	1	2
		1	2	2	2	4	7	9	1
			1	1	1	2	4	7	9

- F F F F F F F F
- # of page faults = 8
- So even though the number of available frames increased, the number of page faults did not decrease.

First-In-First-Out (FIFO) Algorithm

Advantages

- Simple and easy to implement.
- Low overhead.

Disadvantages

- Poor performance.
- Doesn’t consider the frequency of use or last used time, simply replaces the oldest page.
- Suffers from Belady’s Anomaly(i.e. more page faults when we increase the number of page frames).

Least Recently Used (LRU)Algorithm

- Least Recently Used page replacement algorithm keeps track of page usage over a short period of time.
- It works on the idea that the pages that have been most heavily used in the past are most likely to be used heavily in the future too.
- In LRU, whenever page replacement happens, the page which has not been used for the longest amount of time is replaced.

Least Recently Used (LRU)Algorit hm

- Advantages
  - Efficient.
  - Doesn't suffer from Belady’s Anomaly.
- Disadvantages
  - Complex Implementation.
  - Expensive.
  - Requires hardware support.

Pseudocode for LRU

Let us say that s is the main memory's capacity to hold pages and pages is the list containing all pages currently present in the **main memory**.

1. Iterate through the referenced pages.
    1. If the current page is already present in pages:
        1. Remove the current page from pages.
        2. Append the current page to the end of pages.
        3. Increment page hits.
    2. Else:
        1. Increment page faults.
        2. If pages contains fewer pages than its capacity s:
            1. Append current page into pages.
        3. Else:
            1. Remove the first page from pages.
            2. Append the current page at the end of pages.
2. Return the number of page hits and page faults.

Optimal Page Replacement

- Optimal Page Replacement algorithm is the best page replacement algorithm as it gives the least number of page faults.
- It is also known as OPT, clairvoyant replacement algorithm, or Belady's optimal page replacement policy.
- In this algorithm, pages are replaced which would not be used for the longest duration of time in the future, i.e., the pages in the memory which are going to be referred farthest in the future are replaced.

Optimal Page Replacement

- This algorithm was introduced long back and is difficult to implement because it requires future knowledge of the program behavior.
- However, it is possible to implement optimal page replacement on the second run by using the page reference information collected on the first run.

Optimal Page Replacement

- Advantages
  - Easy to Implement.
  - Simple data structures are used.
  - Highly efficient.
- Disadvantages
  - Requires future knowledge of the program.
  - Time-consuming.

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Allocation of frames in OS

- The main memory of the system is divided into frames.
- The OS has to allocate a sufficient number of frames for each process and to do so, the OS uses various algorithms.
- The five major ways to allocate frames are as follows:

1. Proportional frame allocation
2. Priority frame allocation
3. Global replacement allocation
4. Local replacement allocation
5. Equal frame allocation

Allocation of frames in OS

- The proportional frame allocation algorithm allocates frames based on the size that is necessary for the execution and the number of total frames the memory has.
- The only disadvantage of this algorithm is it does not allocate frames based on priority. This situation is solved by Priority frame allocation.

1. Proportional frame allocation

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2. Priority frame allocation

- Priority frame allocation allocates frames based on the priority of the processes and the number of frame allocations.
- If a process is of high priority and needs more frames then the process will be allocated that many frames. The allocation of lower priority processes occurs after it.

3. Global replacement allocation

- When there is a page fault in the operating system, then the global replacement allocation takes care of it.
- The process with lower priority can give frames to the process with higher priority to avoid page faults.

4. Local replacement allocation

- In local replacement allocation, the frames of pages can be stored on the same page.
- It doesn't influence the behavior of the process as it did in global replacement allocation.

5. Equal frame allocation

- In equal frame allocation, the processes are allocated equally among the processes in the operating system.
- The only disadvantage in equal frame allocation is that a process requires more frames for allocation for execution and there are only a set number of frames.

What is Fragmentation?

- Processes are stored and removed from memory, which creates free memory space, which are too small to use by other processes.
- After sometimes, that processes not able to allocate to memory blocks because its small size and memory blocks always remain unused is called fragmentation.
- This type of problem happens during a dynamic memory allocation system when free blocks are quite small, so it is not able to fulfill any request.

What is Fragmentation?

- Two types of Fragmentation methods are:

1. External fragmentation

External fragmentation can be reduced by rearranging memory contents to place all free memory together in a single block.

2. Internal fragmentation

The internal fragmentation can be reduced by assigning the smallest partition, which is still good enough to carry the entire process.

Thrashing in OS (Operating System)

- Thrash is a term used in computer science to describe the poor performance of a virtual memory system when the same pages are loaded repeatedly owing to a shortage of main memory to store them in secondary memory.
- Thrashing happens in computer science when a computer's virtual memory resources are overutilized, resulting in a persistent state of paging and page faults, which inhibits most application-level activity.
- It causes the computer's performance to decline or collapse. The scenario can last indefinitely unless the user stops certain running apps or active processes to free up extra virtual memory resources.

Algorithms during Thrashing

1. Global Page Replacement
2. Local Page Replacement

Algorithms during Thrashing

1. Global Page Replacement
   - Since global page replacement can bring any page, it tries to bring more pages whenever thrashing is found.
   - But what actually will happen is that no process gets enough frames, and as a result, the thrashing will increase more and more.
   - Therefore, the global page replacement algorithm is not suitable when thrashing happens.

Algorithms during Thrashing

2. Local Page Replacement

- Unlike the global page replacement algorithm, local page replacement will select pages which only belong to that process.
- So there is a chance to reduce the thrashing. But it is proven that there are many disadvantages if we use local page replacement.
- Therefore, local page replacement is just an alternative to global page replacement in a thrashing scenario.